MODEL QUESTION PAPER

TED (15) 4121 *Reg. No……….*

Revision (2015) *Signature…….*

FOURTH SEMESTER DIPLOMA EXAMINATION IN WOOD AND PAPER TECHNOLOGY

CHEMICAL ENGINEERING

(Maximum marks: 100) *Time: 3 Hours*

PART-A

(Maximum marks: 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define extraction
2. State adiabatic saturation temp
3. Recall Fourier's law of heat conduction
4. State filter aid
5. Define internal energy

PART-B

(Maximum marks: 30)

II. A

nswer any five of the following questions. Each question carries 6marks.

1. Summarize any five qualities required for solvent for extraction
2. Describe any three rate controlling factors of absorption in two or three sentences.
3. Reproduce Kick's law and Rittinger's law
4. Apply law of conservation of energy and formulate heat balance equation for an evaporator
5. Demonstrate azeotropic distillation
6. Describe Cutting machine (knife cutter) without diagram.
7. Reproduce three modes of heat transfer

PART-C

(Maximum marks: 60)

Answer one full question from each module.

Module-i

III. (a). Define distillation. 2

(b). Reproduce six types of distillation. 9

(c). Predict any four application of distillation. 4

Or

IV. (a). Demonstrate in two or three sentences- loading point flooding point. 4

(b). List any four industrial application of extraction. 4

(c). Explain bubble cap extraction column without diagram. 7

V. (a). Recall humidification. 2

(b). Explain wet bulb theory. 7

(c). Describe three factors influencing wet bulb temperature. 6

VI. (a). In a double pipe heat exchanger hot fluid enters at 100 ^{0} C and leaves at90 ^{0}C and cold fluid enters at 30 ^{0}C and leaves at 50 ^{0}C. Calculate the LMTD for two flow patterns. 8

(b). Explain nucleate boiling and film boiling with a rough graphical representation. 7

VII. (a). Explain the principle and working of a plate and frame filter press with a neat diagram. 8

(b). With the help of a neat diagram explain the working of ball mill. 7

VIII. (a). Draw and explain open circuit grinding. 9

(b). Distinguish between crushing and grinding. 6

IX. (a). Name any four ways of expressing the concentration of a solution. 4

(b). Reproduce mole fraction and mass fraction 4

(c). The mass fraction of KCl in its aqueous solution is 0.051. What is the mole fraction of sodium chloride in the solution? 8

X. (a). Define density and specific gravity. 4

(b). A rock has a volume of 30cm^3 and a mass of 60g. What is its density? 5

(c). A sample of granite with density 2.8 g/cm^3. The density of water is 1.0 g/cm^3. What is the specific gravity of the granite? 6